Wangjie Su

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Research Interests

Current focus: Reliable AI agents for scientific simulation and industrial workflows, with an emphasis on domain-tool and knowledge integration, long-horizon memory and context, multi-agent orchestration, failure recovery, and evaluation.

Education

Peking University, B.S. in Theoretical and Applied Mechanics **Sep 2023 – Present**
Peking University, Double Degree in Intelligent Science and Technology **Sep 2024 – Present**

- **GPA**: 3.818/4.0
- **Selected Coursework**: Data Structures and Algorithms (**100**), Programming in AI (**100**), Computer Vision, Probability and Statistics, Numerical Analysis, Generative Modeling, Intelligent Robotics, Trustworthy ML, LLMs and NLG, and Physics Simulation for Computer Graphics.

Research Experience

AI + Fusion Energy Series Project *Undergraduate Researcher* **Mar 2026 – Present**
Xinao Technology Innovation Center, Peking University | PIs: Prof. Yunfeng Liang and Prof. Songfang Huang

- **Research problem:** Studying *long-horizon research autonomy*: preserving executable research state across interruptions under open-ended goals, delayed feedback, and finite resources, while advancing only through independent evidence.
- **Research framework:** Developing orchestration/execution, evidence/verification, memory/evolution, and governance/collaboration beneath a replaceable research loop; posing ablations on persistence, verifiability, self-healing, accumulation, and governability.
- **Progress: Nexgent** serves as the long-running substrate and can monitor long fusion simulations (background threads that run and watch jobs, check commands for numerical acceptance, recovery after interruptions); migrating concrete fusion tasks is the next stage.

Combustion Laboratory, Peking University *Undergraduate Researcher* **May 2025 – Present**
PI: Prof. Zheng Chen

- **Background:** Studying fundamental combustion problems in energy conversion and propulsion, including flame dynamics, chemical kinetics, transport, ignition, flame propagation and stabilization, and laminar flame speed.
- **Current work:** Exploring mathematical formulations of multiscale combustion processes from conservation laws and reaction mechanisms, together with machine-learning approaches to predicting key quantities and approximating complex processes.
- **Methodological focus:** Comparing the regimes of mechanistic and data-driven models, with attention to physical consistency, interpretability, data efficiency, and out-of-distribution generalization.

Work Experience

Hangzhou Yuanxi Technology Co., Ltd. *Independent Project Lead* **2026.06 – 2026.09**
Early Technical Team Member | System: ManySelves — A Recoverable Local Runtime for File-Defined Agent Teams

- **Capability definition:** Reframed an agent team as a versionable capability specification rather than a fixed application or prompt bundle: Identity contracts define roles, boundaries, I/O, and handoffs; Skill documents encode reusable methods and acceptance criteria; a stable runtime realizes both without domain-specific rewrites.
- **Execution abstraction:** Contributed end to end across the frontend, backend, and runtime; independently designed and implemented the runtime kernel and capability packages. Requests are specialized and compiled into a frozen plan that a stateless kernel advances and persists step by step, with isolated parallel branches that keep failures, nested subworkflows written back step by step, same-run resume, structured-submission

correction, truncated-output continuation, and reuse of completed tool results; users talk only to a Main Agent that coordinates the specialists.

- **Reporting capability:** The bundled power-distribution safety reporting capability runs a nine-stage flow built around a judgment ledger, splitting work so that code handles structure (gap classification, related-item routing, reconciliation, assembly) while the model makes professional judgments that cite records line by line; revisions produce a new version under the same run and reopen only the modules whose facts changed. Two more capability packages cover parameter adjustment and checklist review; FastAPI exposes only conversation and task submission.

Selected Projects

Nexgent: A Desktop Agent Team That Learns *Independent Project* **May 2026 – Present**

- **Two-level organization:** A general-purpose desktop agent (TypeScript monorepo; Electron desktop app and CLI over one in-process SDK) that works in the user's project folder. A coordinator only splits, dispatches, and summarizes; threads work in their own copies of the project and, when a piece is large, use `organize` to assign one to six members with generic roles, merging changes back level by level.
- **Organizational review:** Records the cost, time, failures, and wasted work of every part of each team run, changes only the costliest spot, and keeps or reverts it on the next similar task while keeping a solo control group; experience notes are enabled or disabled by randomized with/without comparisons, and outcomes count only deterministic checks, user corrections, and external review.
- **Reliable execution:** Mid-run user messages take effect on the next model request, and the user's own words survive context compaction verbatim; deliverables are checked automatically and reworked in the same turn on failure; an append-only JSONL session log and usage ledger allow recovery after a hard kill, with OS-level sandboxing, approvals, and three-level cost limits.

Tiny PyTorch: CUDA Deep-Learning Framework *Course Project, Extended* **Fall 2024; Oct 2026**

- **Framework:** Built a PyTorch-style Python package (`tinytensor`: tensor, reverse-mode autodiff, nn / functional / fused layers, optimizers) on hand-written C++/CUDA kernels with pybind11 bindings; the whole training step runs without CPU read-back and is replayed as a CUDA Graph.
- **Kernels:** bf16 tensor-core convolution (mma.sync, cp.async pipelines, split-K), grouped convolution, BatchNorm, LayerNorm, single-kernel attention, and fused Conv-BN-Add-ReLU / Linear-GELU layers (GEMM via cuBLAS); operators, whole-network finite differences, and first training steps are tested against PyTorch.
- **Results:** trained ViT-Tiny, ResNet18, ResNeXt26, and VGG16 on ImageNet-1k 64x64 on one A100 (top-1 56.5% / 56.0% / 62.6% / 57.6%), matching PyTorch epoch by epoch over the first 10 epochs; 2.56x / 1.39x faster per epoch than torch.compile on ViT-Tiny / ResNet18, on par on ResNeXt26, 30% slower on VGG16.

Where Is My Codex (Flow Rescue) *Course Project* **Spring 2026**

- **Physics:** Co-developed a 2D fluid-puzzle game and implemented Position-Based Fluids in Python/Taichi with uniform-grid neighbor search, density constraints, terrain collisions, and GPU/CPU backends.
- **Engineering:** Built five playable levels and a Tiled import pipeline; added headless tests and profiling for digging, collision, channel stability, goal collection, grid construction, and constraint solving.

Honors & Awards

- National Scholarship and Merit Student, Peking University (2025)
- Second Prize, Beijing Division, National College Student Mathematical Modeling Competition (2025)
- Outstanding Learning Award and Third Scholarship, Peking University (2024)
- First Prize, Beijing College Student Mathematics Competition (2024)

Skills

Programming: Python, C/C++, CUDA, TypeScript, MATLAB, LaTeX, Taichi
Tools & Frameworks: PyTorch, Git, CMake, Linux, scikit-learn, Jupyter, OpenFOAM
Languages: Chinese (native); English (CET-6: 613)